Design and Analysis of Computer Algorithms

01:198:344

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Bellman-Ford Algorithm: Overview

- **Purpose**: Computes shortest paths from source s to all other vertices in G.
- Handles graphs with negative edge weights, as long as no negative weight cycles exist.
- **Initialization**:
  - Set $d[s] = 0$ (distance from source to itself is zero).
  - For all $v \neq s$, set $d[v] = +\infty$ (initially assume all other vertices are unreachable).

Bellman-Ford Algorithm: Steps

1. For $n-1$ iterations:
   - For each edge (u, v) with weight $w(u, v)$:
   - If $d[v] > d[u] + w(u, v)$:
     - Update $d[v] = d[u] + w(u, v)$.
   - This "relaxes" edge (u, v) by updating $d[v]$ if a shorter path is found.
2. After $n-1$ iterations, each vertex's distance represents the shortest path from s.

The Bellman-Ford Algorithm: High Level

We maintain:

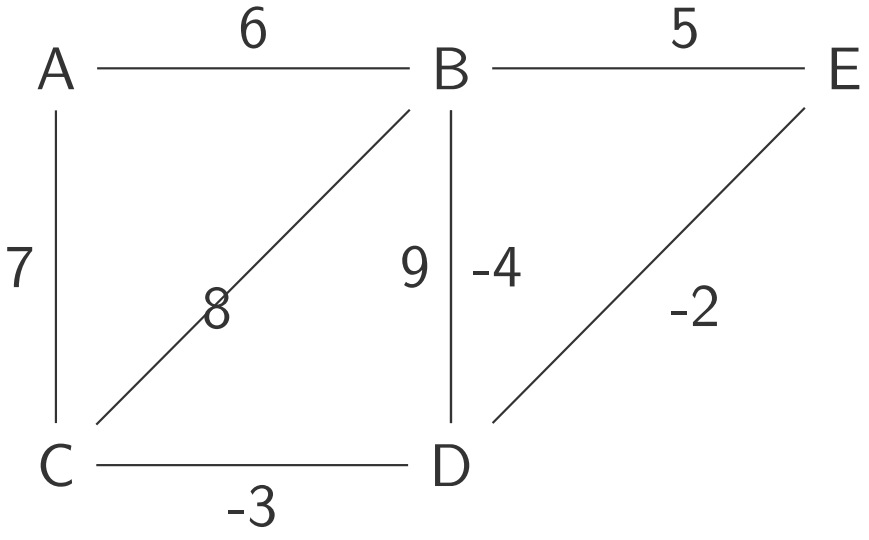
- $d[v]$: Current shortest distance estimate from s to v.
- $\pi[v]$: Predecessor of v in the shortest path found so far.

Steps:

1. **Initialization:**
   - Set $d[s] = 0$.
   - Set $d[v] = \infty$ for all $v \neq s$.
   - Set $\pi[v] = \text{NIL}$ for all v.
2. **Relaxation Passes:** Repeat $V - 1$ times:
   - For each edge $(u, v) \in E$:
   - **Relax** the edge: If $d[u] + w(u, v) < d[v]$, then update:
     - $d[v] = d[u] + w(u, v)$
     - $\pi[v] = u$
3. **Negative Cycle Check:** For each edge $(u, v) \in E$:
   - If $d[u] + w(u, v) < d[v]$, then a negative cycle reachable from s exists. Report it and terminate (or handle as needed).

Example 1: Graph Setup (No Negative Cycle)

Goal: Find shortest paths from source **A**.

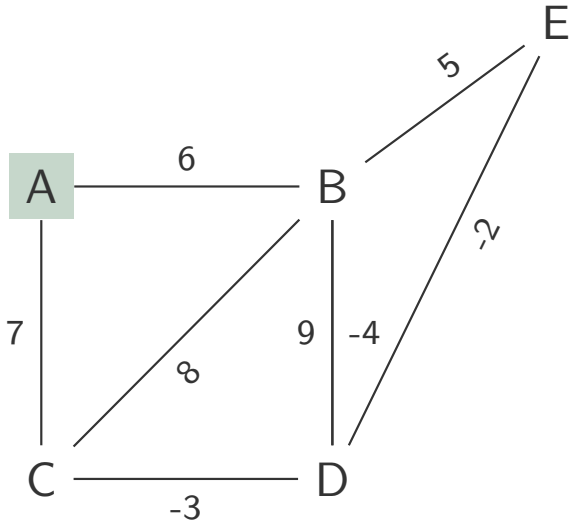


$V = 5$, $E = 8$. We need

$V - 1 = 4$ relaxation passes, plus one check pass.

Example 1: Initialization (Pass 0)

Goal: Set initial distances. $d[A] = 0$, others ∞. Predecessors $\pi[v]$ $=$ NIL.

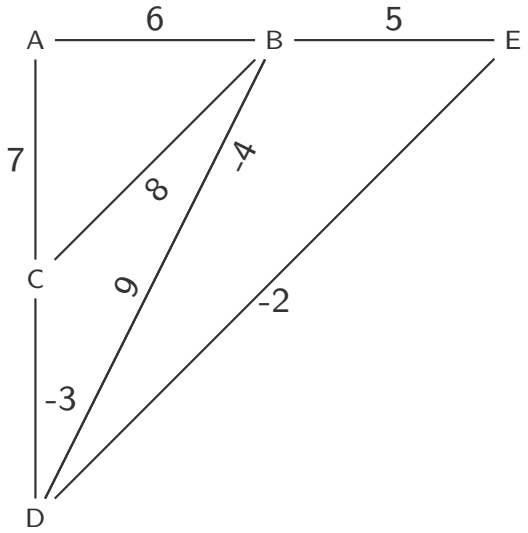


Distances (d) / Predecessors (π)

Vertex	$d[v]$	$\pi[v]$
A	0	NIL
B	∞	NIL
C	∞	NIL
D	∞	NIL
E	∞	NIL

Example 1: After Pass 1

Action: Relax all 8 edges. Edges from A update B and C.



Distances / Predecessors

Vertex	$d[v]$	$\pi[v]$
A	0	NIL
B	6	A
C	7	A
D	∞	NIL
E	∞	NIL

Updates:

Relax (A,B):

$d[A] + w(A, B) = 0 + 6 = 6 < \infty.$

Update $d[B] = 6, \pi[B] = A.$

Relax (A,C):

$d[A] + w(A, C) = 0 + 7 = 7 < \infty.$

Update $d[C] = 7, \pi[C] = A.$

(Other edges relaxed, no updates as source distance is ∞).

Example 1: Setup

Graph Details

- Vertices: A, B, C, D, E $\quad (|V| = 5)$
- Source: A
- Edges (Weight):
  - (A,B): 6, (A,C): 7
  - (B,C): 8, (B,D): -4, (B,E): 5
  - (C,D): -3
  - (D,B): 9
  - (E,D): -2
- $|E| = 8$

Bellman-Ford Parameters

- Passes needed: $|V| - 1 = 4$
- Check pass: 1 (Total 5 passes)
- Relaxation Order (Example): (A,B), (A,C), (B,C), (B,D), (B,E), (C,D), (D,B), (E,D)

Example 1: Initialization Pass 1

Initialization (Pass 0)

- $d = \{A : 0, B : \infty, C : \infty, D : \infty, E : \infty\}$
- $\pi = \{A : \text{NIL}, B : \text{NIL}, C : \text{NIL}, D : \text{NIL}, E : \text{NIL}\}$

Pass 1 Relaxations

- Relax (A,B): $0 + 6 < \infty \implies d[B] = 6, \pi[B] = A$
- Relax (A,C): $0 + 7 < \infty \implies d[C] = 7, \pi[C] = A$
- Relax (B,C): $6 + 8 \not< 7$
- Relax (B,D): $6 - 4 < \infty \implies d[D] = 2, \pi[D] = B$
- Relax (B,E): $6 + 5 < \infty \implies d[E] = 11, \pi[E] = B$
- Relax (C,D): $7 - 3 = 4 \not< d[D] = 2$
- Relax (D,B): $2 + 9 \not< d[B] = 6$
- Relax (E,D): $11 - 2 = 9 \not< d[D] = 2$

End of Pass 1 State

- $d = \{A : 0, B : 6, C : 7, D : 2, E : 11\}$
- $\pi = \{A : \text{NIL}, B : A, C : A, D : B, E : B\}$

Example 1: Passes 2, 3, 4

Pass 2 Relaxations Checking all edges with Pass 1 values:

- (A,B): $0 + 6 \not< 6$
- (A,C): $0 + 7 \not< 7$
- (B,C): $6 + 8 \not< 7$
- (B,D): $6 - 4 \not< 2$
- (B,E): $6 + 5 \not< 11$
- (C,D): $7 - 3 \not< 2$
- (D,B): $2 + 9 \not< 6$
- (E,D): $11 - 2 \not< 2$

Result: No updates occurred in Pass 2. Pass 3 ($|V| - 2$)

- Since Pass 2 had no updates, Pass 3 will also have **no updates**.

Pass 4 ($|V| - 1$)

- Since Pass 3 had no updates, Pass 4 will also have **no updates**.

End of Pass 4 State (Final Candidate Distances)

- $d = \{A : 0, B : 6, C : 7, D : 2, E : 11\}$
- $\pi = \{A : \text{NIL}, B : A, C : A, D : B, E : B\}$

Example 1: Pass 5 (Negative Cycle Check)

Action: Relax all edges using final candidate distances from Pass 4.

- Check (A,B): $0 + 6 \not< 6$
- Check (A,C): $0 + 7 \not< 7$
- Check (B,C): $6 + 8 \not< 7$
- Check (B,D): $6 - 4 \not< 2$
- Check (B,E): $6 + 5 \not< 11$
- Check (C,D): $7 - 3 = 4 \not< 2$
- Check (D,B): $2 + 9 = 11 \not< 6$
- Check (E,D): $11 - 2 = 9 \not< 2$

Result

- No relaxation improved any distance $d[v]$.
- No negative-weight cycle reachable from source A detected.

Example 1: Conclusion

Final Shortest Paths The Bellman-Ford algorithm terminated successfully without detecting negative cycles reachable from A. The shortest path distances and predecessors found after Pass 4 are:

- $d[A] = 0$, $\pi[A] = \text{NIL}$ (Path: A)
- $d[B] = 6$, $\pi[B] = A$ (Path: A $\rightarrow$ B)
- $d[C] = 7$, $\pi[C] = A$ (Path: A $\rightarrow$ C)
- $d[D] = 2$, $\pi[D] = B$ (Path: A $\rightarrow$ B $\rightarrow$ D)
- $d[E] = 11$, $\pi[E] = B$ (Path: A $\rightarrow$ B $\rightarrow$ E)

Example 2: Setup

Graph Details

- Vertices: A, B, C, D $\quad (|V| = 4)$
- Source: A
- Edges (Weight): (A,B): 5, (B,C): 2, (C,D): -5, (D,B): -2
- $|E| = 4$

Bellman-Ford Parameters

- Passes needed: $|V| - 1 = 3$
- Check pass: 1 (Total 4 passes)
- Relaxation Order (Example): (A,B), (B,C), (C,D), (D,B)

Note: Cycle B $\rightarrow$ C $\rightarrow$ D $\rightarrow$ B has weight $2 + (-5) + (-2) = -5$.

Example 2: Initialization Pass 1

Initialization (Pass 0)

- $d = \{A : 0, B : \infty, C : \infty, D : \infty\}$
- $\pi = \{A : \text{NIL}, B : \text{NIL}, C : \text{NIL}, D : \text{NIL}\}$

Pass 1 Relaxations

- Relax (A,B): $0 + 5 < \infty \implies d[B] = 5, \pi[B] = A$
- Relax (B,C): $5 + 2 < \infty \implies d[C] = 7, \pi[C] = B$
- Relax (C,D): $7 - 5 < \infty \implies d[D] = 2, \pi[D] = C$
- Relax (D,B): $2 - 2 = 0 < d[B] = 5 \implies d[B] = 0, \pi[B] = D$

End of Pass 1 State

- $d = \{A : 0, B : 0, C : 7, D : 2\}$
- $\pi = \{A : \text{NIL}, B : D, C : B, D : C\}$

Example 2: Pass 2

Pass 2 Relaxations (Using values from end of Pass 1)

- Relax (A,B): $0 + 5 \not< d[B] = 0$
- Relax (B,C): $0 + 2 = 2 < d[C] = 7 \implies d[C] = 2, \pi[C] = B$
- Relax (C,D): $2 - 5 = -3 < d[D] = 2 \implies d[D] = -3, \pi[D] = C$
- Relax (D,B): $-3 - 2 = -5 < d[B] = 0 \implies d[B] = -5, \pi[B] = D$

End of Pass 2 State

- $d = \{A : 0, B : -5, C : 2, D : -3\}$
- $\pi = \{A : \text{NIL}, B : D, C : B, D : C\}$

Example 2: Pass 3 ($|V| - 1$ Pass)

Pass 3 Relaxations (Using values from end of Pass 2)

- Relax (A,B): $0 + 5 \not< d[B] = -5$
- Relax (B,C): $-5 + 2 = -3 < d[C] = 2 \implies d[C] = -3, \pi[C] = B$
- Relax (C,D): $-3 - 5 = -8 < d[D] = -3 \implies d[D] = -8, \pi[D] = C$
- Relax (D,B): $-8 - 2 = -10 < d[B] = -5 \implies d[B] = -10, \pi[B] = D$

End of Pass 3 State

- $d = \{A : 0, B : -10, C : -3, D : -8\}$
- $\pi = \{A : \text{NIL}, B : D, C : B, D : C\}$

Example 2: Pass 4 (Negative Cycle Check)

Action: Relax all edges using final candidate distances from Pass 3.

- Check (A,B): $0 + 5 = 5 \not< d[B] = -10$
- Check (B,C): $d[B] + 2 = -10 + 2 = -8 < d[C] = -3$.
  - Update Detected! ($d[C]$ could be improved to -8).
- Check (C,D): (Further checks not strictly needed once one update is found)
- Check (D,B): (Further checks not strictly needed once one update is found)

Result

- An edge relaxation (B,C) resulted in an improvement during the $|V|$-th pass.
- This signals a Negative-weight cycle reachable from source A.

Example 2

Outcome

- The algorithm detected that distances could still be improved in Pass 4 (the check pass).
- This occurs because the negative cycle B $\rightarrow$ C $\rightarrow$ D $\rightarrow$ B (weight -5) is reachable from A.
- Traversing the cycle allows path weights to decrease indefinitely.
- Therefore, shortest paths are not well-defined for nodes B, C, and D in this graph.
- Bellman-Ford successfully reports the existence of such a cycle.

Bellman-Ford: Runtime Complexity

- Initialization: $O(V)$
- Core Relaxation Loops: $V - 1$ passes, each relaxing E edges ($O(1)$/edge) $\implies O(VE)$ total.
- Negative Cycle Check: One additional pass relaxing E edges $\implies O(E)$.

Overall Runtime: The dominant term is $O(VE)$, so the total complexity is

$$\boldsymbol{O(VE)}$$

- Handles negative edge weights.
- Detects reachable negative cycles.
- Slower than Dijkstra's on graphs without negative edges.

Proof of Correctness for Bellman-Ford

- **Goal**: Show $d[v] = \text{dist}(s, v)$ for all $v \in V$.
- Proof has two main parts:
  1. $d[v] \geq \text{dist}(s, v)$: $d[v]$ never underestimates shortest path distance.
  2. $d[v] \leq \text{dist}(s, v)$: $d[v]$ eventually reaches the actual shortest path distance.

$d[v] \geq \text{dist}(s, v)$

- For each v, $d[v]$ is the total weight along a path from s to v.
- Since $\text{dist}(s, v)$ is defined as the minimum weight path, we have $d[v] \geq \text{dist}(s, v)$.

$d[v] \leq \text{dist}(s, v)$

- **Inductive Setup**:
  - Define $\text{dist}_i(s, v)$: minimum path weight from s to v using at most i edges.
  - Base case ($i = 0$): $d[s] = 0$ and $d[v] = +\infty$ for all $v \neq s$.
- **Inductive Step**: Assume for i-th iteration that $d[v] \leq \text{dist}_i(s, v)$.
  - For any $s \to v$ path P with $\leq i + 1$ edges, let u be the last vertex before v in P.
  - By induction, $d[u] \leq \text{dist}_i(s, u)$.
  - Update $d[v] = \min(d[v], d[u] + w_{uv})$, ensuring $d[v] \leq \text{dist}_{i+1}(s, v)$.
- After $n - 1$ iterations, $d[v] \leq \text{dist}(s, v)$.

Bellman-Ford Algorithm: Negative Weights

Negative Weight Handling:

- Algorithm works with negative weights if no negative cycles exist.
- Negative cycles can be detected if further updates occur in an n-th iteration.
- The algorithm still correctly return the shortest distance for vertices not connected to the negative cycle.
- If we can

Bellman-Ford Algorithm: Runtime Analysis

- **Each Iteration**: $O(m)$ time to scan all edges.
- **Total Time Complexity**: $O(n \cdot m)$, since algorithm iterates up to $n - 1$ times.
- **Summary**:
  - Suitable for graphs with negative weights.
  - Slower than Dijkstra's when all edge weights are positive.

Bellman-Ford: Sequential vs Parallel Updates

Sequential Edge-Relaxation Updates:

- Iteratively relaxes edges $|V| - 1$ times.
- For each edge (u, v):

$$d[v] = \min(d[v], d[u] + w(u, v))$$

- Updates are applied immediately, allowing new values to propagate within the same iteration.

Parallel Updates:

- Computes $d_i(v)$, the distance from s to v with at most i edges, for all vertices in parallel for each i (iteration):

$$d_i(v) = \min(d_{i-1}(s, v), d_{i-1}(u) + w(u, v))$$

- Uses values only from $i - 1$, ensuring no propagation within the same iteration.
- Maintains separate states for each i.

Bellman-Ford: Runtime Complexity

Core Loop:

- The algorithm performs $V - 1$ passes.
- In each pass, it iterates through and relaxes *all* E edges.
- Relaxation takes $O(1)$ time per edge.
- Cost of main loops: $(V - 1) \times O(E) = O(VE)$.

Negative Cycle Check:

- Requires one additional pass over all E edges.
- Cost: $O(E)$.

Initialization:

- Setting initial distances takes $O(V)$.

Overall Runtime:

- Total $= O(V) + O(VE) + O(E) = \boldsymbol{O(VE)}$.
- Slower than Dijkstra's for graphs with non-negative weights, especially sparse graphs.
- Main advantage: Ability to handle negative edge weights and detect negative cycles.

Cheapest Flights with K Stops

Problem Statement:

- **Input:**
  - n cities, list of flights (edges) as (source, destination, cost)
  - Starting city src, destination city dst, maximum stops K
- **Goal:** Find the minimum cost to travel from src to dst with at most K stops.

Cheapest Flights with K Stops

Problem Statement:

- **Input:**
  - n cities, list of flights (edges) as (source, destination, cost)
  - Starting city src, destination city dst, maximum stops K
- **Goal:** Find the minimum cost to travel from src to dst with at most K stops.

Solution: Bellman-Ford Approach

1. Initialize distance array dist with:

$$\text{dist[src]} = 0, \quad \text{dist[v]} = \infty \text{ for all other } v$$

2. For each iteration from 1 to $K + 1$ (representing stops):
   - Create a copy of dist as temp_dist to store updates for this iteration.
   - For each flight (u, v, cost), update:

$$\text{temp_dist[v]} = \min(\text{temp_dist[v]}, \text{dist[u]} + \text{cost})$$

   - Set dist = temp_dist at the end of the iteration.
3. Return dist[dst] if reachable, otherwise return -1.

Floyd-Warshall Algorithm Overview

- **Goal**: Compute shortest paths between all pairs of vertices.
- **Works with**: Positive and negative edge weights (no negative cycles).
- **Technique**: Dynamic programming via incremental path updates.
- **Output**: Distance matrix $\text{dist}[i][j]$ = shortest path length from i to j.
- **Complexity**:
  - Time: $O(n^3)$
  - Space: $O(n^2)$

Floyd-Warshall: Dynamic Programming Insight

Key Idea: $d_k[i][j]$ the shortest distance from i to j using $1, 2, \cdots k$ as intermediate vertices.

Recurrence:

$$d_k[i][j] = \min\left(d_{k-1}[i][j],\ d_{k-1}[i][k] + d_{k-1}[k][j]\right)$$

Explanation:

- Either the shortest path from i to j avoids k,
- Or it goes through k, in which case we combine:

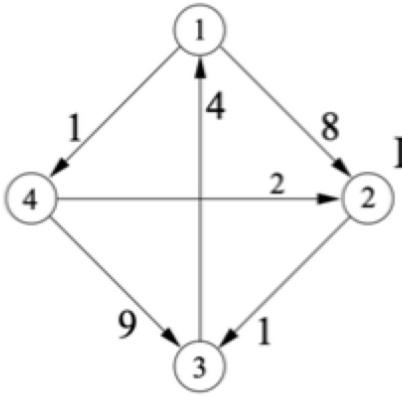
$$i \to k \quad + \quad k \to j$$

Floyd-Warshall: Pseudocode

1. Initialize distance matrix dist:
   - $\text{dist}[i][j] = w(i, j)$ if edge exists
   - $\text{dist}[i][j] = \infty$ otherwise
   - $\text{dist}[i][i] = 0$
2. Loop over all intermediate vertices k:
   - For all i, j:

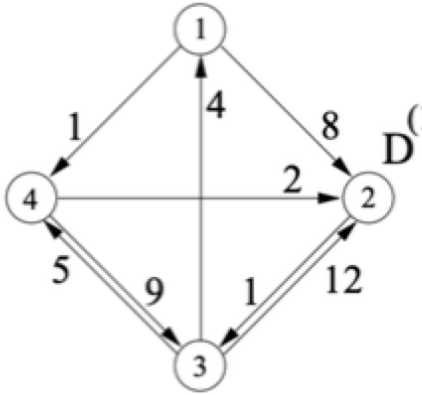
$$\text{dist}[i][j] \leftarrow \min(\text{dist}[i][j],\ \text{dist}[i][k] + \text{dist}[k][j])$$

Example

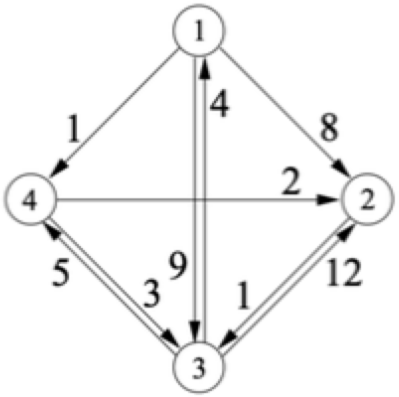


$$D^{(0)} = \begin{bmatrix} 0 & 8 & ? & 1 \\ ? & 0 & 1 & ? \\ 4 & ? & 0 & ? \\ ? & 2 & 9 & 0 \end{bmatrix}$$

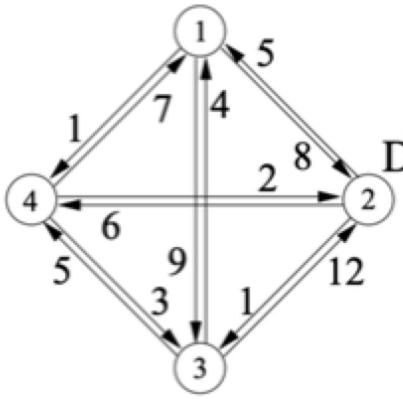
? = infinity



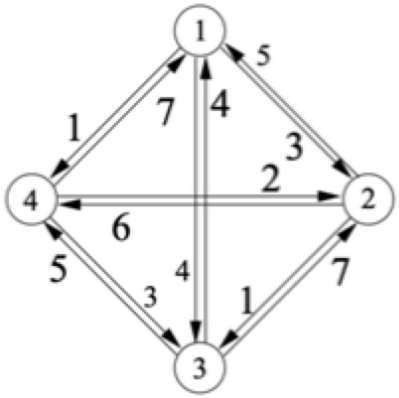
$$D^{(1)} = \begin{bmatrix} 0 & 8 & ? & 1 \\ ? & 0 & 1 & ? \\ 4 & 12 & 0 & 5 \\ ? & 2 & 9 & 0 \end{bmatrix}$$



$$D^{(2)} = \begin{bmatrix} 0 & 8 & 9 & 1 \\ ? & 0 & 1 & ? \\ 4 & 12 & 0 & 5 \\ ? & 2 & 3 & 0 \end{bmatrix}$$



$$D^{(3)} = \begin{bmatrix} 0 & 8 & 9 & 1 \\ 5 & 0 & 1 & 6 \\ 4 & 12 & 0 & 5 \\ 7 & 2 & 3 & 0 \end{bmatrix}$$



$$D^{(4)} = \begin{bmatrix} 0 & 3 & 4 & 1 \\ 5 & 0 & 1 & 6 \\ 4 & 7 & 0 & 5 \\ 7 & 2 & 3 & 0 \end{bmatrix}$$

Sequential vs Parallel Updates

Sequential Updates (Floyd-Warshall):

- In-place updates: $\text{dist}[i][j]$ is updated immediately.
- Allows values to propagate within a single iteration over k.
- Memory efficient: only one distance matrix.

Parallel Updates (Layered DP):

- Compute $d_k[i][j]$ using only d_{k-1}:

$$d_k[i][j] = \min(d_{k-1}[i][j],\ d_{k-1}[i][k] + d_{k-1}[k][j])$$

- Each iteration uses a fresh matrix; updates do not affect same round.
- Easier to parallelize; more memory-intensive.

Negative Cycle Detection

- Floyd-Warshall can detect negative cycles.
- **Key Observation:** If $\text{dist}[i][i] < 0$ after the algorithm:
  - A negative-weight cycle exists that is reachable from i.
  - Shortest paths are undefined for vertices involved in such cycles.

Path Reconstruction

- Maintain a predecessor matrix $\text{pred}[i][j]$:
  - Initially: $\text{pred}[i][j] = i$ if edge (i, j) exists, else null.
  - When updating $\text{dist}[i][j]$ via k, set:

$$\text{pred}[i][j] = \text{pred}[k][j]$$

- **To reconstruct path from i to j:**
  - Follow the chain: $j \leftarrow \text{pred}[i][j] \leftarrow \cdots \leftarrow i$

Floyd-Warshall: Complexity Summary

- **Time Complexity:** $O(n^3)$
  - Triple nested loop over all vertex pairs and intermediates.
- **Space Complexity:** $O(n^2)$
  - Distance matrix for all pairs.
  - Optional: Predecessor matrix for path reconstruction.
- **Use Case:** Small or dense graphs; all-pairs shortest paths; graphs with negative weights.

City with the few close neighbours

Problem:

- Given n cities connected by weighted edges, find the city with the smallest number of neighbors within a given distance threshold.
- If multiple cities have the same number of reachable neighbors, return the city with the highest ID.

Approach:

1. Use Floyd-Warshall algorithm to compute all-pairs shortest paths between cities.
2. For each city i, count the number of cities j with distance$(i, j) \leq$ threshold.
3. Track the city with the smallest neighbor count; if tied, choose the city with the highest ID.